Matteo Zandi

On general relativity:

python computations.

February 11, 2024

matteo.zandi2@studio.unibo.it

Contents

Page

Contents **ii**

1 Exterior Schwarschild metric **3**
1.1 Christoffel symbols 3
1.2 Riemann tensor 4
1.3 $(0,2)$ Ricci tensor 6
1.4 $(1,1)$ Ricci tensor 6
1.5 Ricci scalar 6
1.6 $(0,2)$ Einstein tensor 7
1.7 $(1,1)$ Einstein tensor 7

2 Interior Schwarschild metric **9**
2.1 Christoffel symbols 9
2.2 Riemann tensor 10
2.3 $(0,2)$ Ricci tensor 11
2.4 $(1,1)$ Ricci tensor 12
2.5 Ricci scalar 12
2.6 $(0,2)$ Einstein tensor 12
2.7 $(1,1)$ Einstein tensor 13

3 FRW metric **15**
3.1 Christoffel symbols 15
3.2 Riemann tensor 17
3.3 $(0,2)$ Ricci tensor 18
3.4 $(1,1)$ Ricci tensor 19
3.5 Ricci scalar 19
3.6 $(0,2)$ Einstein tensor 19
3.7 $(1,1)$ Einstein tensor 20

Abstract

In these notes, we will compute general relativity with the help of python. We will take the exterior Schwarschild metric, the interior Schwarschild metric and the Friedmann-Robertson-Walker metric. We will calculate the Christoffel symbols, the Riemann tensor, the Ricci tensor and the Einstein tensor.

Chapter 1

Exterior Schwarschild metric

The metric is

$$g = \begin{bmatrix} -A(r) & 0 & 0 & 0 \\ 0 & B(r) & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2(\theta) \end{bmatrix} ,$$

.

Its inverse is

$$g = \begin{bmatrix} -\frac{1}{A(r)} & 0 & 0 & 0 \\ 0 & \frac{1}{B(r)} & 0 & 0 \\ 0 & 0 & \frac{1}{r^2} & 0 \\ 0 & 0 & 0 & \frac{1}{r^2 \sin^2(\theta)} \end{bmatrix} ,$$

.

1.1 Christoffel symbols

The non-vanishing components of the Christoffel symbols are

$$\Gamma^0_{01} = \frac{0.5 \frac{d}{dr} A(r)}{A(r)} ,$$

$$\Gamma^0_{10} = \frac{0.5 \frac{d}{dr} A(r)}{A(r)} ,$$

$$\Gamma^1_{00} = \frac{0.5 \frac{d}{dr} A(r)}{B(r)} ,$$

$$\Gamma^1_{11} = \frac{0.5 \frac{d}{dr} B(r)}{B(r)} ,$$

$$\Gamma^1_{22} = -\frac{1.0 r}{B(r)} ,$$

$$\Gamma^1_{33} = -\frac{1.0r\sin^2(\theta)}{B(r)} \ ,$$

$$\Gamma^2_{12} = \frac{1.0}{r} \ ,$$

$$\Gamma^2_{21} = \frac{1.0}{r} \ ,$$

$$\Gamma^2_{33} = -0.5\sin(2\theta) \ ,$$

$$\Gamma^3_{13} = \frac{1.0}{r} \ ,$$

$$\Gamma^3_{23} = \frac{1.0}{\tan(\theta)} \ ,$$

$$\Gamma^3_{31} = \frac{1.0}{r} \ ,$$

$$\Gamma^3_{32} = \frac{1.0}{\tan(\theta)} \ ,$$

.

1.2 Riemann tensor

The non-vanishing components of the Riemann tensor are

$$R^0{}_{101} = -\frac{0.5\frac{d^2}{dr^2}A(r)}{A(r)} + \frac{0.25\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{A(r)B(r)} + \frac{0.25\left(\frac{d}{dr}A(r)\right)^2}{A^2(r)} \ ,$$

$$R^0{}_{110} = \frac{0.5\frac{d^2}{dr^2}A(r)}{A(r)} - \frac{0.25\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{A(r)B(r)} - \frac{0.25\left(\frac{d}{dr}A(r)\right)^2}{A^2(r)} \ ,$$

$$R^0{}_{202} = -\frac{0.5r\frac{d}{dr}A(r)}{A(r)B(r)} \ ,$$

$$R^0{}_{220} = \frac{0.5r\frac{d}{dr}A(r)}{A(r)B(r)} \ ,$$

$$R^0{}_{303} = -\frac{0.5r\sin^2(\theta)\frac{d}{dr}A(r)}{A(r)B(r)} \ ,$$

$$R^0{}_{330} = \frac{0.5r\sin^2(\theta)\frac{d}{dr}A(r)}{A(r)B(r)} \ ,$$

$$R^1{}_{001} = -\frac{0.5\frac{d^2}{dr^2}A(r)}{B(r)} + \frac{0.25\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{B^2(r)} + \frac{0.25\left(\frac{d}{dr}A(r)\right)^2}{A(r)B(r)} \ ,$$

$$R^1{}_{010} = \frac{0.5\frac{d^2}{dr^2}A(r)}{B(r)} - \frac{0.25\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{B^2(r)} - \frac{0.25\left(\frac{d}{dr}A(r)\right)^2}{A(r)B(r)} \ ,$$

$$R^1{}_{212} = \frac{0.5r\frac{d}{dr}B(r)}{B^2(r)} \ ,$$

$$R^1{}_{221} = -\frac{0.5r\frac{d}{dr}B(r)}{B^2(r)} \ ,$$

$$R^1{}_{313} = \frac{0.5r\sin^2(\theta)\frac{d}{dr}B(r)}{B^2(r)} \ ,$$

$$R^1{}_{331} = -\frac{0.5r\sin^2(\theta)\frac{d}{dr}B(r)}{B^2(r)} \ ,$$

$$R^2{}_{002} = -\frac{0.5\frac{d}{dr}A(r)}{rB(r)} \ ,$$

$$R^2{}_{020} = \frac{0.5\frac{d}{dr}A(r)}{rB(r)} \ ,$$

$$R^2{}_{112} = -\frac{0.5\frac{d}{dr}B(r)}{rB(r)} \ ,$$

$$R^2{}_{121} = \frac{0.5\frac{d}{dr}B(r)}{rB(r)} \ ,$$

$$R^2{}_{323} = \frac{1.0\,(B(r)-1)\sin^2(\theta)}{B(r)} \ ,$$

$$R^2{}_{332} = \frac{1.0\cdot(1-B(r))\sin^2(\theta)}{B(r)} \ ,$$

$$R^3{}_{003} = -\frac{0.5\frac{d}{dr}A(r)}{rB(r)} \ ,$$

$$R^3{}_{030} = \frac{0.5\frac{d}{dr}A(r)}{rB(r)} \ ,$$

$$R^3{}_{113} = -\frac{0.5\frac{d}{dr}B(r)}{rB(r)} \ ,$$

$$R^3{}_{131} = \frac{0.5\frac{d}{dr}B(r)}{rB(r)} \ ,$$

$$R^3{}_{223} = -1.0 + \frac{1.0}{B(r)} \ ,$$

$$R^3{}_{232} = 1.0 - \frac{1.0}{B(r)} \ ,$$

.

1.3 $(0,2)$ Ricci tensor

The non-vanishing components of the $(0,2)$ Ricci tensor are

$$R_{00} = \frac{0.5\frac{d^2}{dr^2}A(r)}{B(r)} - \frac{0.25\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{B^2(r)} - \frac{0.25\left(\frac{d}{dr}A(r)\right)^2}{A(r)B(r)} + \frac{1.0\frac{d}{dr}A(r)}{rB(r)} ,$$

$$R_{11} = -\frac{0.5\frac{d^2}{dr^2}A(r)}{A(r)} + \frac{0.25\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{A(r)B(r)} + \frac{0.25\left(\frac{d}{dr}A(r)\right)^2}{A^2(r)} + \frac{1.0\frac{d}{dr}B(r)}{rB(r)} ,$$

$$R_{22} = \frac{0.5r\frac{d}{dr}B(r)}{B^2(r)} - \frac{0.5r\frac{d}{dr}A(r)}{A(r)B(r)} + 1.0 - \frac{1.0}{B(r)} ,$$

$$R_{33} = \frac{\left(0.5rA(r)\frac{d}{dr}B(r) - 0.5rB(r)\frac{d}{dr}A(r) + 1.0A(r)B^2(r) - 1.0A(r)B(r)\right)\sin^2(\theta)}{A(r)B^2(r)} ,$$

.

1.4 $(1,1)$ Ricci tensor

The non-vanishing components of the $(1,1)$ Ricci tensor are

$$R^0{}_0 = -\frac{0.5\frac{d^2}{dr^2}A(r)}{A(r)B(r)} + \frac{0.25\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{A(r)B^2(r)} + \frac{0.25\left(\frac{d}{dr}A(r)\right)^2}{A^2(r)B(r)} - \frac{1.0\frac{d}{dr}A(r)}{rA(r)B(r)} ,$$

$$R^1{}_1 = -\frac{0.5\frac{d^2}{dr^2}A(r)}{A(r)B(r)} + \frac{0.25\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{A(r)B^2(r)} + \frac{0.25\left(\frac{d}{dr}A(r)\right)^2}{A^2(r)B(r)} + \frac{1.0\frac{d}{dr}B(r)}{rB^2(r)} ,$$

$$R^2{}_2 = \frac{0.5\frac{d}{dr}B(r)}{rB^2(r)} - \frac{0.5\frac{d}{dr}A(r)}{rA(r)B(r)} + \frac{1.0}{r^2} - \frac{1.0}{r^2B(r)} ,$$

$$R^3{}_3 = \frac{0.5\frac{d}{dr}B(r)}{rB^2(r)} - \frac{0.5\frac{d}{dr}A(r)}{rA(r)B(r)} + \frac{1.0}{r^2} - \frac{1.0}{r^2B(r)} ,$$

.

1.5 Ricci scalar

The Ricci scalar is

$$R = -\frac{1.0\frac{d^2}{dr^2}A(r)}{A(r)B(r)} + \frac{0.5\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{A(r)B^2(r)} + \frac{0.5\left(\frac{d}{dr}A(r)\right)^2}{A^2(r)B(r)} + \frac{2.0\frac{d}{dr}B(r)}{rB^2(r)} - \frac{2.0\frac{d}{dr}A(r)}{rA(r)B(r)} + \frac{2.0}{r^2} - \frac{2.0}{r^2B(r)} .$$

1.6 (0, 2) Einstein tensor

The non-vanishing components of the (0, 2) Einstein tensor are

$$G_{00} = \frac{1.0\left(r\frac{d}{dr}B(r) + B^2(r) - B(r)\right)A(r)}{r^2B^2(r)} ,$$

$$G_{11} = \frac{1.0\frac{d}{dr}A(r)}{rA(r)} - \frac{1.0B(r)}{r^2} + \frac{1.0}{r^2} ,$$

$$G_{22} = \frac{1.0r\left(0.5rA(r)B(r)\frac{d^2}{dr^2}A(r) - 0.25rA(r)\frac{d}{dr}A(r)\frac{d}{dr}B(r) - 0.25rB(r)\left(\frac{d}{dr}A(r)\right)^2 - 0.5A^2(r)\frac{d}{dr}B(r\right.}{A^2(r)B^2(r)}$$

$$G_{33} = \frac{1.0r\left(0.5rA(r)B(r)\frac{d^2}{dr^2}A(r) - 0.25rA(r)\frac{d}{dr}A(r)\frac{d}{dr}B(r) - 0.25rB(r)\left(\frac{d}{dr}A(r)\right)^2 - 0.5A^2(r)\frac{d}{dr}B(r\right.}{A^2(r)B^2(r)}$$

.

1.7 (1, 1) Einstein tensor

The non-vanishing components of the (1, 1) Einstein tensor are

$$G^0{}_0 = -\frac{1.0\frac{d}{dr}B(r)}{rB^2(r)} - \frac{1.0}{r^2} + \frac{1.0}{r^2B(r)} ,$$

$$G^1{}_1 = \frac{1.0\left(r\frac{d}{dr}A(r) + (1 - B(r))A(r)\right)}{r^2A(r)B(r)} ,$$

$$G^2{}_2 = \frac{0.5\frac{d^2}{dr^2}A(r)}{A(r)B(r)} - \frac{0.25\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{A(r)B^2(r)} - \frac{0.25\left(\frac{d}{dr}A(r)\right)^2}{A^2(r)B(r)} - \frac{0.5\frac{d}{dr}B(r)}{rB^2(r)} + \frac{0.5\frac{d}{dr}A(r)}{rA(r)B(r)} ,$$

$$G^3{}_3 = \frac{0.5\frac{d^2}{dr^2}A(r)}{A(r)B(r)} - \frac{0.25\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{A(r)B^2(r)} - \frac{0.25\left(\frac{d}{dr}A(r)\right)^2}{A^2(r)B(r)} - \frac{0.5\frac{d}{dr}B(r)}{rB^2(r)} + \frac{0.5\frac{d}{dr}A(r)}{rA(r)B(r)} ,$$

.

Chapter 2

Interior Schwarschild metric

The metric is

$$g = \begin{bmatrix} -e^{\nu(r)} & 0 & 0 & 0 \\ 0 & e^{\lambda(r)} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2(\theta) \end{bmatrix} ,$$

.

Its inverse is

$$g = \begin{bmatrix} -e^{-\nu(r)} & 0 & 0 & 0 \\ 0 & e^{-\lambda(r)} & 0 & 0 \\ 0 & 0 & \frac{1}{r^2} & 0 \\ 0 & 0 & 0 & \frac{1}{r^2 \sin^2(\theta)} \end{bmatrix} ,$$

.

2.1 Christoffel symbols

The non-vanishing components of the Christoffel symbols are

$$\Gamma^0_{01} = 0.5 \frac{d}{dr} \nu(r) ,$$

$$\Gamma^0_{10} = 0.5 \frac{d}{dr} \nu(r) ,$$

$$\Gamma^1_{00} = 0.5 e^{-\lambda(r)+\nu(r)} \frac{d}{dr} \nu(r) ,$$

$$\Gamma^1_{11} = 0.5 \frac{d}{dr} \lambda(r) ,$$

$$\Gamma^1_{22} = -1.0 r e^{-\lambda(r)} ,$$

$$\Gamma^1_{33} = -1.0 r e^{-\lambda(r)} \sin^2(\theta) ,$$

$$\Gamma^2_{12} = \frac{1.0}{r} \ ,$$
$$\Gamma^2_{21} = \frac{1.0}{r} \ ,$$
$$\Gamma^2_{33} = -0.5 \sin(2\theta) \ ,$$
$$\Gamma^3_{13} = \frac{1.0}{r} \ ,$$
$$\Gamma^3_{23} = \frac{1.0}{\tan(\theta)} \ ,$$
$$\Gamma^3_{31} = \frac{1.0}{r} \ ,$$
$$\Gamma^3_{32} = \frac{1.0}{\tan(\theta)} \ ,$$
.

2.2 Riemann tensor

The non-vanishing components of the Riemann tensor are

$$R^0{}_{101} = 0.25\frac{d}{dr}\lambda(r)\frac{d}{dr}\nu(r) - 0.25\left(\frac{d}{dr}\nu(r)\right)^2 - 0.5\frac{d^2}{dr^2}\nu(r) \ ,$$

$$R^0{}_{110} = -0.25\frac{d}{dr}\lambda(r)\frac{d}{dr}\nu(r) + 0.25\left(\frac{d}{dr}\nu(r)\right)^2 + 0.5\frac{d^2}{dr^2}\nu(r) \ ,$$

$$R^0{}_{202} = -0.5re^{-\lambda(r)}\frac{d}{dr}\nu(r) \ ,$$

$$R^0{}_{220} = 0.5re^{-\lambda(r)}\frac{d}{dr}\nu(r) \ ,$$

$$R^0{}_{303} = -0.5re^{-\lambda(r)}\sin^2(\theta)\frac{d}{dr}\nu(r) \ ,$$

$$R^0{}_{330} = 0.5re^{-\lambda(r)}\sin^2(\theta)\frac{d}{dr}\nu(r) \ ,$$

$$R^1{}_{001} = \left(0.25\frac{d}{dr}\lambda(r)\frac{d}{dr}\nu(r) - 0.25\left(\frac{d}{dr}\nu(r)\right)^2 - 0.5\frac{d^2}{dr^2}\nu(r)\right)e^{-\lambda(r)+\nu(r)} \ ,$$

$$R^1{}_{010} = \left(-0.25\frac{d}{dr}\lambda(r)\frac{d}{dr}\nu(r) + 0.25\left(\frac{d}{dr}\nu(r)\right)^2 + 0.5\frac{d^2}{dr^2}\nu(r)\right)e^{-\lambda(r)+\nu(r)} \ ,$$

$$R^1{}_{212} = 0.5re^{-\lambda(r)}\frac{d}{dr}\lambda(r) \ ,$$

$$R^1{}_{221} = -0.5re^{-\lambda(r)}\frac{d}{dr}\lambda(r) \ ,$$
$$R^1{}_{313} = 0.5re^{-\lambda(r)}\sin^2(\theta)\frac{d}{dr}\lambda(r) \ ,$$
$$R^1{}_{331} = -0.5re^{-\lambda(r)}\sin^2(\theta)\frac{d}{dr}\lambda(r) \ ,$$
$$R^2{}_{002} = -\frac{0.5e^{-\lambda(r)+\nu(r)}\frac{d}{dr}\nu(r)}{r} \ ,$$
$$R^2{}_{020} = \frac{0.5e^{-\lambda(r)+\nu(r)}\frac{d}{dr}\nu(r)}{r} \ ,$$
$$R^2{}_{112} = -\frac{0.5\frac{d}{dr}\lambda(r)}{r} \ ,$$
$$R^2{}_{121} = \frac{0.5\frac{d}{dr}\lambda(r)}{r} \ ,$$
$$R^2{}_{323} = 1.0\left(e^{\lambda(r)}-1\right)e^{-\lambda(r)}\sin^2(\theta) \ ,$$
$$R^2{}_{332} = 1.0\cdot\left(1-e^{\lambda(r)}\right)e^{-\lambda(r)}\sin^2(\theta) \ ,$$
$$R^3{}_{003} = -\frac{0.5e^{-\lambda(r)+\nu(r)}\frac{d}{dr}\nu(r)}{r} \ ,$$
$$R^3{}_{030} = \frac{0.5e^{-\lambda(r)+\nu(r)}\frac{d}{dr}\nu(r)}{r} \ ,$$
$$R^3{}_{113} = -\frac{0.5\frac{d}{dr}\lambda(r)}{r} \ ,$$
$$R^3{}_{131} = \frac{0.5\frac{d}{dr}\lambda(r)}{r} \ ,$$
$$R^3{}_{223} = -1.0 + 1.0e^{-\lambda(r)} \ ,$$
$$R^3{}_{232} = 1.0 - 1.0e^{-\lambda(r)} \ ,$$
$$.$$

2.3 (0, 2) Ricci tensor

The non-vanishing components of the (0, 2) Ricci tensor are

$$R_{00} = \frac{\left(-0.25r\frac{d}{dr}\lambda(r)\frac{d}{dr}\nu(r) + 0.25r\left(\frac{d}{dr}\nu(r)\right)^2 + 0.5r\frac{d^2}{dr^2}\nu(r) + 1.0\frac{d}{dr}\nu(r)\right)e^{-\lambda(r)+\nu(r)}}{r} \ ,$$

$$R_{11} = 0.25\frac{d}{dr}\lambda(r)\frac{d}{dr}\nu(r) - 0.25\left(\frac{d}{dr}\nu(r)\right)^2 - 0.5\frac{d^2}{dr^2}\nu(r) + \frac{1.0\frac{d}{dr}\lambda(r)}{r} \ ,$$

$$R_{22} = \left(0.5r\left(\frac{d}{dr}\lambda(r) - \frac{d}{dr}\nu(r)\right) + 1.0e^{\lambda(r)} - 1.0\right)e^{-\lambda(r)} \ ,$$

$$R_{33} = \left(0.5r\frac{d}{dr}\lambda(r) - 0.5r\frac{d}{dr}\nu(r) + 1.0e^{\lambda(r)} - 1.0\right)e^{-\lambda(r)}\sin^2(\theta) \ ,$$

.

2.4 (1, 1) Ricci tensor

The non-vanishing components of the (1, 1) Ricci tensor are

$$R^0{}_0 = \frac{\left(0.25r\frac{d}{dr}\lambda(r)\frac{d}{dr}\nu(r) - 0.25r\left(\frac{d}{dr}\nu(r)\right)^2 - 0.5r\frac{d^2}{dr^2}\nu(r) - 1.0\frac{d}{dr}\nu(r)\right)e^{-\lambda(r)}}{r} \ ,$$

$$R^1{}_1 = \frac{\left(r\left(0.25\frac{d}{dr}\lambda(r)\frac{d}{dr}\nu(r) - 0.25\left(\frac{d}{dr}\nu(r)\right)^2 - 0.5\frac{d^2}{dr^2}\nu(r)\right) + 1.0\frac{d}{dr}\lambda(r)\right)e^{-\lambda(r)}}{r} \ ,$$

$$R^2{}_2 = \frac{\left(0.5r\left(\frac{d}{dr}\lambda(r) - \frac{d}{dr}\nu(r)\right) + 1.0e^{\lambda(r)} - 1.0\right)e^{-\lambda(r)}}{r^2} \ ,$$

$$R^3{}_3 = \frac{\left(0.5r\frac{d}{dr}\lambda(r) - 0.5r\frac{d}{dr}\nu(r) + 1.0e^{\lambda(r)} - 1.0\right)e^{-\lambda(r)}}{r^2} \ ,$$

.

2.5 Ricci scalar

The Ricci scalar is

$$R = \frac{\left(0.5r^2\frac{d}{dr}\lambda(r)\frac{d}{dr}\nu(r) - 0.5r^2\left(\frac{d}{dr}\nu(r)\right)^2 - 1.0r^2\frac{d^2}{dr^2}\nu(r) + 2.0r\frac{d}{dr}\lambda(r) - 2.0r\frac{d}{dr}\nu(r) + 2.0e^{\lambda(r)} - 2.0\right)}{r^2}$$

2.6 (0, 2) Einstein tensor

The non-vanishing components of the (0, 2) Einstein tensor are

$$G_{00} = \frac{1.0\left(r\frac{d}{dr}\lambda(r) + e^{\lambda(r)} - 1\right)e^{-\lambda(r)+\nu(r)}}{r^2} \ ,$$

$$G_{11} = \frac{1.0\left(r\frac{d}{dr}\nu(r) - e^{\lambda(r)} + 1\right)}{r^2} \ ,$$

$$G_{22} = r\left(-0.25r\frac{d}{dr}\lambda(r)\frac{d}{dr}\nu(r) + 0.25r\left(\frac{d}{dr}\nu(r)\right)^2 + 0.5r\frac{d^2}{dr^2}\nu(r) - 0.5\frac{d}{dr}\lambda(r) + 0.5\frac{d}{dr}\nu(r)\right)e^{-\lambda(r)} \ ,$$

$$G_{33} = 1.0r\left(-0.25r\frac{d}{dr}\lambda(r)\frac{d}{dr}\nu(r) + 0.25r\left(\frac{d}{dr}\nu(r)\right)^2 + 0.5r\frac{d^2}{dr^2}\nu(r) - 0.5\frac{d}{dr}\lambda(r) + 0.5\frac{d}{dr}\nu(r)\right)e^{-\lambda(}$$

.

2.7 (1, 1) Einstein tensor

The non-vanishing components of the (1, 1) Einstein tensor are

$$G^0{}_0 = \frac{1.0\left(-r\frac{d}{dr}\lambda(r) - e^{\lambda(r)} + 1\right)e^{-\lambda(r)}}{r^2},$$

$$G^1{}_1 = \frac{1.0\left(r\frac{d}{dr}\nu(r) - e^{\lambda(r)} + 1\right)e^{-\lambda(r)}}{r^2},$$

$$G^2{}_2 = \frac{\left(-0.25r\frac{d}{dr}\lambda(r)\frac{d}{dr}\nu(r) + 0.25r\left(\frac{d}{dr}\nu(r)\right)^2 + 0.5r\frac{d^2}{dr^2}\nu(r) - 0.5\frac{d}{dr}\lambda(r) + 0.5\frac{d}{dr}\nu(r)\right)e^{-\lambda(r)}}{r},$$

$$G^3{}_3 = \frac{\left(-0.25r\frac{d}{dr}\lambda(r)\frac{d}{dr}\nu(r) + 0.25r\left(\frac{d}{dr}\nu(r)\right)^2 + 0.5r\frac{d^2}{dr^2}\nu(r) - 0.5\frac{d}{dr}\lambda(r) + 0.5\frac{d}{dr}\nu(r)\right)e^{-\lambda(r)}}{r},$$

.

Chapter 3

FRW metric

The metric is

$$g = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -\frac{a^2(t)}{kr^2-1} & 0 & 0 \\ 0 & 0 & r^2a^2(t) & 0 \\ 0 & 0 & 0 & r^2a^2(t)\sin^2(\theta) \end{bmatrix} ,$$

.

Its inverse is

$$g = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & \frac{-kr^2+1}{a^2(t)} & 0 & 0 \\ 0 & 0 & \frac{1}{r^2a^2(t)} & 0 \\ 0 & 0 & 0 & \frac{1}{r^2a^2(t)\sin^2(\theta)} \end{bmatrix} ,$$

.

3.1 Christoffel symbols

The non-vanishing components of the Christoffel symbols are

$$\Gamma^0_{11} = -\frac{1.0a(t)\frac{d}{dt}a(t)}{kr^2-1} ,$$

$$\Gamma^0_{22} = 1.0r^2a(t)\frac{d}{dt}a(t) ,$$

$$\Gamma^0_{33} = 1.0r^2a(t)\sin^2(\theta)\frac{d}{dt}a(t) ,$$

$$\Gamma^1_{01} = \frac{1.0\frac{d}{dt}a(t)}{a(t)} ,$$

$$\Gamma^1_{10} = \frac{1.0\frac{d}{dt}a(t)}{a(t)} \ ,$$

$$\Gamma^1_{11} = -\frac{1.0kr}{kr^2 - 1} \ ,$$

$$\Gamma^1_{22} = 1.0r\left(kr^2 - 1\right) \ ,$$

$$\Gamma^1_{33} = 1.0r\left(kr^2 - 1\right)\sin^2(\theta) \ ,$$

$$\Gamma^2_{02} = \frac{1.0\frac{d}{dt}a(t)}{a(t)} \ ,$$

$$\Gamma^2_{12} = \frac{1.0}{r} \ ,$$

$$\Gamma^2_{20} = \frac{1.0\frac{d}{dt}a(t)}{a(t)} \ ,$$

$$\Gamma^2_{21} = \frac{1.0}{r} \ ,$$

$$\Gamma^2_{33} = -0.5\sin(2\theta) \ ,$$

$$\Gamma^3_{03} = \frac{1.0\frac{d}{dt}a(t)}{a(t)} \ ,$$

$$\Gamma^3_{13} = \frac{1.0}{r} \ ,$$

$$\Gamma^3_{23} = \frac{1.0}{\tan(\theta)} \ ,$$

$$\Gamma^3_{30} = \frac{1.0\frac{d}{dt}a(t)}{a(t)} \ ,$$

$$\Gamma^3_{31} = \frac{1.0}{r} \ ,$$

$$\Gamma^3_{32} = \frac{1.0}{\tan(\theta)} \ ,$$

.

3.2 Riemann tensor

The non-vanishing components of the Riemann tensor are

$$R^0{}_{101} = -\frac{1.0a(t)\frac{d^2}{dt^2}a(t)}{kr^2-1} \ ,$$

$$R^0{}_{110} = \frac{1.0a(t)\frac{d^2}{dt^2}a(t)}{kr^2-1} \ ,$$

$$R^0{}_{202} = 1.0r^2a(t)\frac{d^2}{dt^2}a(t) \ ,$$

$$R^0{}_{220} = -1.0r^2a(t)\frac{d^2}{dt^2}a(t) \ ,$$

$$R^0{}_{303} = 1.0r^2a(t)\sin^2(\theta)\frac{d^2}{dt^2}a(t) \ ,$$

$$R^0{}_{330} = -1.0r^2a(t)\sin^2(\theta)\frac{d^2}{dt^2}a(t) \ ,$$

$$R^1{}_{001} = \frac{1.0\frac{d^2}{dt^2}a(t)}{a(t)} \ ,$$

$$R^1{}_{010} = -\frac{1.0\frac{d^2}{dt^2}a(t)}{a(t)} \ ,$$

$$R^1{}_{212} = 1.0r^2\left(k+\left(\frac{d}{dt}a(t)\right)^2\right) \ ,$$

$$R^1{}_{221} = 1.0r^2\left(-k-\left(\frac{d}{dt}a(t)\right)^2\right) \ ,$$

$$R^1{}_{313} = 1.0r^2\left(k+\left(\frac{d}{dt}a(t)\right)^2\right)\sin^2(\theta) \ ,$$

$$R^1{}_{331} = 1.0r^2\left(-k-\left(\frac{d}{dt}a(t)\right)^2\right)\sin^2(\theta) \ ,$$

$$R^2{}_{002} = \frac{1.0\frac{d^2}{dt^2}a(t)}{a(t)} \ ,$$

$$R^2{}_{020} = -\frac{1.0\frac{d^2}{dt^2}a(t)}{a(t)} \ ,$$

$$R^2{}_{112} = \frac{1.0\left(k+\left(\frac{d}{dt}a(t)\right)^2\right)}{kr^2-1} \ ,$$

$$R^2{}_{121} = \frac{1.0\left(-k - \left(\frac{d}{dt}a(t)\right)^2\right)}{kr^2 - 1} \ ,$$

$$R^2{}_{323} = 1.0r^2\left(k + \left(\frac{d}{dt}a(t)\right)^2\right)\sin^2(\theta) \ ,$$

$$R^2{}_{332} = -1.0r^2\left(k + \left(\frac{d}{dt}a(t)\right)^2\right)\sin^2(\theta) \ ,$$

$$R^3{}_{003} = \frac{1.0\frac{d^2}{dt^2}a(t)}{a(t)} \ ,$$

$$R^3{}_{030} = -\frac{1.0\frac{d^2}{dt^2}a(t)}{a(t)} \ ,$$

$$R^3{}_{113} = \frac{1.0\left(k + \left(\frac{d}{dt}a(t)\right)^2\right)}{kr^2 - 1} \ ,$$

$$R^3{}_{131} = \frac{1.0\left(-k - \left(\frac{d}{dt}a(t)\right)^2\right)}{kr^2 - 1} \ ,$$

$$R^3{}_{223} = 1.0r^2\left(-k - \left(\frac{d}{dt}a(t)\right)^2\right) \ ,$$

$$R^3{}_{232} = 1.0r^2\left(k + \left(\frac{d}{dt}a(t)\right)^2\right) \ ,$$

.

3.3 $(0, 2)$ Ricci tensor

The non-vanishing components of the $(0, 2)$ Ricci tensor are

$$R_{00} = -\frac{3.0\frac{d^2}{dt^2}a(t)}{a(t)} \ ,$$

$$R_{11} = \frac{-2.0k - 1.0a(t)\frac{d^2}{dt^2}a(t) - 2.0\left(\frac{d}{dt}a(t)\right)^2}{kr^2 - 1} \ ,$$

$$R_{22} = r^2 \cdot \left(2.0k + 1.0a(t)\frac{d^2}{dt^2}a(t) + 2.0\left(\frac{d}{dt}a(t)\right)^2\right) \ ,$$

$$R_{33} = r^2 \cdot \left(2.0k + 1.0a(t)\frac{d^2}{dt^2}a(t) + 2.0\left(\frac{d}{dt}a(t)\right)^2\right)\sin^2(\theta) \ ,$$

.

3.4 (1, 1) Ricci tensor

The non-vanishing components of the (1, 1) Ricci tensor are

$$R^0{}_0 = \frac{3.0\frac{d^2}{dt^2}a(t)}{a(t)} \ ,$$

$$R^1{}_1 = \frac{2.0k + 1.0a(t)\frac{d^2}{dt^2}a(t) + 2.0\left(\frac{d}{dt}a(t)\right)^2}{a^2(t)} \ ,$$

$$R^2{}_2 = \frac{2.0k + 1.0a(t)\frac{d^2}{dt^2}a(t) + 2.0\left(\frac{d}{dt}a(t)\right)^2}{a^2(t)} \ ,$$

$$R^3{}_3 = \frac{2.0k + 1.0a(t)\frac{d^2}{dt^2}a(t) + 2.0\left(\frac{d}{dt}a(t)\right)^2}{a^2(t)} \ ,$$

.

3.5 Ricci scalar

The Ricci scalar is

$$R = \frac{6.0\left(k + a(t)\frac{d^2}{dt^2}a(t) + \left(\frac{d}{dt}a(t)\right)^2\right)}{a^2(t)} \ .$$

3.6 (0, 2) Einstein tensor

The non-vanishing components of the (0, 2) Einstein tensor are

$$G_{00} = \frac{3.0\left(k + \left(\frac{d}{dt}a(t)\right)^2\right)}{a^2(t)} \ ,$$

$$G_{11} = \frac{1.0k + 2.0a(t)\frac{d^2}{dt^2}a(t) + 1.0\left(\frac{d}{dt}a(t)\right)^2}{kr^2 - 1} \ ,$$

$$G_{22} = r^2\left(-1.0k - 2.0a(t)\frac{d^2}{dt^2}a(t) - 1.0\left(\frac{d}{dt}a(t)\right)^2\right) \ ,$$

$$G_{33} = r^2\left(-1.0k - 2.0a(t)\frac{d^2}{dt^2}a(t) - 1.0\left(\frac{d}{dt}a(t)\right)^2\right)\sin^2(\theta) \ ,$$

.

3.7 $(1,1)$ Einstein tensor

The non-vanishing components of the $(1,1)$ Einstein tensor are

$$G^{0}{}_{0} = \frac{3.0\left(-k - \left(\frac{d}{dt}a(t)\right)^{2}\right)}{a^{2}(t)} \ ,$$

$$G^{1}{}_{1} = \frac{-1.0k - 2.0a(t)\frac{d^{2}}{dt^{2}}a(t) - 1.0\left(\frac{d}{dt}a(t)\right)^{2}}{a^{2}(t)} \ ,$$

$$G^{2}{}_{2} = \frac{-1.0k - 2.0a(t)\frac{d^{2}}{dt^{2}}a(t) - 1.0\left(\frac{d}{dt}a(t)\right)^{2}}{a^{2}(t)} \ ,$$

$$G^{3}{}_{3} = \frac{-1.0k - 2.0a(t)\frac{d^{2}}{dt^{2}}a(t) - 1.0\left(\frac{d}{dt}a(t)\right)^{2}}{a^{2}(t)} \ ,$$

.

Bibliography

[1] https://github.com/PhysicsZandi/GeneralRelativity/blob/main/source/py/python_gr.tex.